

BEARTOOTH ISLAND, SK, Canada

WMO: 711340

Lat: 59.208N

Lon: 109.693W

Elev: 238

StdP: 98.50

Time Zone: -6.00 (W06)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.5	(c) -34.5	(d) -41.3	(e) 0.1	(f) -36.5	(g) -38.9	(h) 0.1	(i) -34.6	(j) 15.6	(k) -20.9	(l) 13.9	(m) -21.8	(n) 3.6	(o) 270	(p) 0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.3	(c) 23.7	(d) 18.5	(e) 22.0	(f) 17.3	(g) 20.3	(h) 16.2	(i) 18.9	(j) 22.2	(k) 17.6	(l) 20.7	(m) 16.5	(n) 19.4	(o) 3.0	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.9	21.2	16.2	11.9	19.8	15.0	11.0	18.8	54.8	22.4	50.7	20.8	47.2	19.5	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.9	(b) 11.2	(c) 9.9	(e) -37.9	(f) 26.5	(g) 2.3	(h) 3.8	(i) -39.5	(j) 29.3	(k) -40.8	(l) 31.5	(m) -42.1	(n) 33.7	(o) -43.8	(p) 36.4		
(4) 12.9	(5) 11.2	(6) 9.9	(7) -38.3	(8) 20.1	(9) 2.4	(10) 1.8	(11) -40.0	(12) 21.4	(13) -41.4	(14) 22.4	(15) -42.8	(16) 23.4	(17) -44.5	(18) 24.7	(4) 12.9	(5) 11.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-22.9	-17.6	-12.0	-2.9	3.6	10.4	15.8	15.3	9.5	2.1	-7.3	-17.5
	DBStd	14.16	7.69	7.51	7.86	5.69	3.24	3.59	3.17	3.40	3.66	4.39	5.97	8.73
	HDD10.0	4771	1019	774	683	388	198	36	1	3	53	244	520	852
	HDD18.3	7400	1277	1007	941	638	456	239	91	103	266	502	770	1111
	CDD10.0	437	0	0	0	0	1	48	182	169	37	0	0	0
	CDD18.3	25	0	0	0	0	0	2	13	10	0	0	0	0
	CDH23.3	65	0	0	0	0	0	7	32	26	1	0	0	0
	CDH26.7	5	0	0	0	0	0	1	3	2	0	0	0	0
(14) Wind	WSAvg	4.8	4.8	4.7	4.8	4.8	4.9	4.4	3.8	4.0	5.0	5.8	5.5	5.3
(15) Precipitation	PrecAvg	454	23	20	25	24	33	57	69	64	55	35	26	23
	PrecMax	563	48	49	69	63	63	87	116	126	145	63	54	52
	PrecMin	351	9	6	3	1	13	35	38	24	20	11	9	9
	PrecStd	66	10	10	13	12	13	16	20	23	23	15	12	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	1.8	4.8	9.1	16.6	24.0	26.0	25.9	20.7	11.7	3.5	0.8
		MCWB	-1.4	0.6	2.7	5.0	9.9	14.2	18.2	20.2	14.7	8.9	2.0	0.0
	2%	DB	-6.3	-2.1	2.7	6.9	12.7	20.5	24.0	23.7	17.6	9.6	2.3	-1.3
		MCWB	-6.8	-2.9	1.2	4.0	8.2	13.9	18.5	19.1	13.9	7.7	1.5	-1.8
	5%	DB	-10.0	-4.7	0.7	5.4	10.0	18.0	22.3	21.8	15.7	8.3	1.3	-3.9
		MCWB	-10.4	-5.2	-0.3	3.1	6.8	13.1	17.5	17.5	12.8	6.7	0.6	-4.4
	10%	DB	-12.4	-7.1	-1.3	4.0	8.0	15.9	20.7	20.1	14.3	7.1	0.1	-6.3
		MCWB	-12.8	-7.5	-2.2	2.3	5.4	11.9	16.7	16.4	11.8	5.7	-0.4	-6.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	0.7	2.9	5.3	10.9	16.0	20.2	20.9	15.3	9.3	2.2	-0.1
		MCDB	-0.5	1.9	4.4	8.3	14.4	21.1	23.6	24.1	19.1	11.1	2.7	0.4
	2%	WB	-6.2	-3.1	1.4	4.1	8.4	14.2	18.8	19.4	14.0	8.0	1.5	-2.4
		MCDB	-5.8	-2.5	2.6	6.4	11.9	17.3	22.1	22.6	16.8	9.5	2.1	-1.9
	5%	WB	-10.0	-5.7	-0.1	3.1	6.6	13.1	17.8	18.0	12.8	6.9	0.5	-5.0
		MCDB	-9.7	-5.2	0.8	5.0	9.4	16.1	21.1	20.9	14.9	8.3	1.1	-4.5
	10%	WB	-12.6	-8.6	-2.1	2.2	5.4	11.7	16.7	16.9	11.9	6.0	-0.8	-7.3
		MCDB	-12.4	-8.2	-1.2	3.8	7.5	14.8	19.8	19.5	13.8	7.0	-0.3	-7.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	8.3	10.7	10.0	7.7	8.2	7.3	6.5	5.5	4.1	4.2	5.9
		MCDBR	9.2	9.8	10.9	10.1	11.2	11.4	9.2	8.4	7.4	5.7	4.1	7.9
		MCWBR	9.1	10.3	9.5	8.1	7.4	6.8	5.4	5.1	5.3	4.7	4.0	8.6
	5% WB	MCDBR	9.7	10.9	10.6	9.8	10.3	10.1	8.5	7.9	6.8	5.7	4.3	8.3
		MCWBR	9.4	10.3	9.6	7.8	7.3	6.6	5.4	5.2	5.0	4.6	4.0	8.2
(40) Clear-Sky Solar Irradiance	taub		0.207	0.237	0.267	0.314	0.332	0.361	0.385	0.346	0.312	0.278	0.222	0.183
	taud		2.202	2.301	2.308	2.363	2.468	2.387	2.377	2.461	2.580	2.533	2.247	2.089
	Ebn at Noon		654	816	886	894	900	872	839	852	837	765	619	540
	Edh at Noon		41	64	88	101	98	108	107	92	69	53	39	30
(44) All-Sky Solar Radiation	RadAvg		0.41	1.22	2.83	4.54	5.55	6.08	5.50	4.31	2.77	1.27	0.43	0.23
	RadStd		0.05	0.11	0.18	0.31	0.49	0.43	0.35	0.24	0.26	0.14	0.06	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page